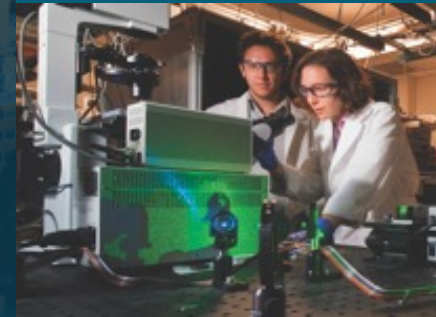


Stanford ME469: Splitting and Stabilization Errors



PRESENTED BY

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Computational Thermal and Fluid Mechanics

Sandia National Laboratories SAND2018-4536 PE



- Block Matrix and Operator Form
- Approximate Factorization
- Splitting Errors
- Stabilization Errors
- Detailed Code Verification
- Conclusions

Standard Incompressible Equation Set

- Consider the uniform density, low-Mach equation system:

$$\frac{\partial u_j}{\partial x_j} = 0$$
$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} = -\frac{\partial p_i}{\partial x_i} + S_i$$

4 Standard Incompressible Equation Set



- Consider the uniform density, low-Mach equation system:

$$\frac{\partial u_j}{\partial x_j} = 0$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} = -\frac{\partial p_i}{\partial x_i} + S_i$$

Time Step Count: 8 Current Time: 0.32

dtN: 0.04 dtNm1: 0.04 gammas: 1.5 -2 0.5

Max Courant: 0.22364 Max Reynolds: 1294.89 (realm_1)

Realm Nonlinear Iteration: 1/1

realm_1::advance_time_step()

NLI	Name	Linear Iter	Linear Res	NLinear Res	Scaled NLR
---	----	-----	-----	-----	
1/2	Equation System Iteration				
1/1	myLowMach				
	MomentumEQS	3	2.62985e-06	0.00890841	1
	ContinuityEQS	4	1.06209e-06	0.0021837	1
2/2	Equation System Iteration				
1/1	myLowMach				
	MomentumEQS	3	2.28651e-08	4.62336e-05	0.00518989
	ContinuityEQS	5	5.2886e-09	5.42362e-05	0.0248369

In our simulation studies, we have been solving this system sequentially and within a nonlinear loop:

```
For ( non-linear iteration ) {
    1. Momentum
    2. Continuity
    If ( converged ) break;
}
```

Today, we learn more about the consequence of this strategy

Introduction to Block Matrix Form

- Consider the divergence of a vector, $\mathbf{DF}$ (a scalar)

$$\frac{\partial F_j}{\partial x_j} = 0 \quad \left\{ \begin{array}{l} \int \frac{\partial F_j}{\partial x_j} dV = 0 \\ \int w \frac{\partial F_j}{\partial x_j} d\Omega = 0 \end{array} \right. \quad \begin{array}{l} \xrightarrow{\text{(Gauss-Divergence)}} \\ \xrightarrow{\text{(Integration-by-parts)}} \end{array} \quad \begin{array}{l} \int F_j n_j dS = 0 \\ - \int F_j \frac{\partial w}{\partial x_j} d\Omega + \int w F_j n_j d\Gamma = 0 \end{array} \quad \begin{array}{l} \xleftarrow{\text{(piecewise-const } w)} \\ \xleftarrow{\text{(piecewise-const } w)} \end{array}$$

- Gradient of a scalar, $\mathbf{Gp}$ (a vector)

$$\frac{\partial p}{\partial x_i} = 0 \quad \left\{ \begin{array}{l} \int \frac{\partial p}{\partial x_i} dV = 0 \\ \int w \frac{\partial p}{\partial x_i} d\Omega = 0 \end{array} \right. \quad \begin{array}{l} \xrightarrow{\text{(Green-Gauss)}} \\ \xrightarrow{\text{(Integration-by-parts)}} \end{array} \quad \begin{array}{l} \int p n_i dS = 0 \\ - \int p \frac{\partial w}{\partial x_i} d\Omega + \int w p n_i d\Gamma = 0 \end{array} \quad \begin{array}{l} \xleftarrow{\text{(piecewise-const } w)} \\ \xleftarrow{\text{(piecewise-const } w)} \end{array}$$

$$\begin{array}{l} \frac{\partial q_j}{\partial x_j} = 0 \\ q_j = -\lambda \frac{\partial T}{\partial x_j} \end{array} \quad \left\{ \begin{array}{l} \int \frac{\partial q_j}{\partial x_j} dV = 0 \\ \int w \frac{\partial q_j}{\partial x_j} d\Omega = 0 \end{array} \right. \quad \begin{array}{l} \xrightarrow{\text{(Gauss-Divergence)}} \\ \xrightarrow{\text{(Integration-by-parts)}} \end{array} \quad \begin{array}{l} - \int \lambda \frac{\partial T}{\partial x_j} n_j dS = 0 \\ - \int q_j \frac{\partial w}{\partial x_j} d\Omega + \int w q_j n_j d\Gamma = 0 \end{array} \quad \begin{array}{l} \xleftarrow{\text{(piecewise-const } w)} \\ \xleftarrow{\text{(piecewise-const } w)} \end{array}$$

6 Equally Valid Pressure Projection Derivations



- Semi-discrete approach (uniform density)

$$\begin{aligned} \rho \frac{\hat{u}_i - u_i^n}{\Delta t} + \frac{\partial}{\partial x_j} (\rho \hat{u}_i u_j^n) &= -\frac{\partial p^n}{\partial x_i} + \frac{\partial \hat{\tau}_{ij}}{\partial x_j} \\ \rho \frac{u_i^{n+1} - \hat{u}_i}{\Delta t} &= -\frac{\partial}{\partial x_i} (p^{n+1} - p^n) \\ \frac{\partial^2}{\partial x_i^2} (p^{n+1} - p^n) &= \frac{\rho}{\Delta t} \frac{\partial \hat{u}_i}{\partial x_i} \end{aligned}$$

- Chorin (1968)
- Time scale is the time step:

$$u_i^{n+1} = \hat{u}_i - \Delta t \left(\frac{1}{\rho} \frac{\partial}{\partial x_i} (p^{n+1} - p^n) \right)$$

- Fully discrete approach

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \mathbf{U}^{n+1} \\ \mathbf{P}^{n+1} \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ \mathbf{0} \end{Bmatrix}$$

- Factored matrix:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & \mathbf{0} \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{D} & -\mathbf{D}\bar{\mathbf{A}}^{-1}\mathbf{G} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \bar{\mathbf{A}}^{-1}\mathbf{G} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$$

$\bar{\mathbf{A}}^{-1}$ is an approximation to $\mathbf{A}^{-1}$

- SIMPLE family of methods
- Time scale is the characteristic scale of : $\bar{\mathbf{A}}^{-1}$

$$\mathbf{U}^{n+1} = \hat{\mathbf{U}} - \bar{\mathbf{A}}^{-1} \mathbf{G} \mathbf{P}^{n+1}$$

- For convection-dominated flows, this looks like $\Delta \mathbf{x} / \mathbf{U}$

7 Introduction to Block Matrix Form for Fluids



- Consider the monolithic, uniform density, low-Mach equation system:

$$\frac{\partial u_j}{\partial x_j} = 0$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} = -\frac{\partial p}{\partial x_i} + S_i$$

that can be written in block form as:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- We seek to factorize this system via:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & \mathbf{B}_1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{B}_2 \mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & \mathbf{A} \mathbf{B}_2 \mathbf{G} \\ \mathbf{D} & (\mathbf{B}_1 + \mathbf{D} \mathbf{B}_2 \mathbf{G}) \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

the exact factorization can be recovered by defining:

$$\left\{ \begin{array}{l} \mathbf{B}_2 = \mathbf{A}^{-1} \\ \mathbf{B}_1 = -\mathbf{D} \mathbf{B}_2 \mathbf{G} \end{array} \right.$$

- $\mathbf{B}_2$ determines the projection time scale, ideally chosen to approximate $\mathbf{A}^{-1}$
- $\mathbf{B}_1$ controls the projection error, ideally chosen to cancel $\mathbf{B} \mathbf{D}_2 \mathbf{G}$



- The approximate factorization

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & \mathbf{B}_1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{B}_2 \mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & \mathbf{A} \mathbf{B}_2 \mathbf{G} \\ \mathbf{D} & (\mathbf{B}_1 + \mathbf{D} \mathbf{B}_2 \mathbf{G}) \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- can be written now as two segregated steps:

$$\text{Momentum and Continuity: } \begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & \mathbf{B}_1 \end{bmatrix} \begin{bmatrix} \hat{u} \\ \hat{p} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} \quad \left\{ \begin{array}{l} \mathbf{A} \hat{u} = \mathbf{f} \\ \mathbf{D} \hat{u} + \mathbf{B}_1 \hat{p} = 0 \end{array} \right.$$

$$\text{Nodal Projection: } \begin{bmatrix} \mathbf{I} & \mathbf{B}_2 \mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{u} \\ \hat{p} \end{bmatrix} \quad \left\{ \begin{array}{l} u^{n+1} = \hat{u} - \mathbf{B}_2 \mathbf{G} p^{n+1} = 0 \\ p^{n+1} = \hat{p} \end{array} \right.$$

- This approach seems to be straight forward, however, what errors have we introduced by this procedure of splitting the monolithic (fully coupled) system?

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} (\mathbf{I} - \mathbf{A} \mathbf{B}_2) \mathbf{G} p^{n+1} \\ -(\mathbf{B}_1 + \mathbf{D} \mathbf{B}_2 \mathbf{G}) p^{n+1} \end{bmatrix} \quad \text{Exact iff: } \begin{array}{l} \mathbf{B}_2 = \mathbf{A}^{-1} \\ \mathbf{B}_1 = -\mathbf{D} \mathbf{B}_2 \mathbf{G} \end{array}$$

- In most cases, $\mathbf{B}_2$ is approximately $\mathbf{A}^{-1}$ and a first-order temporal splitting error is noted

9 Recall Mesh Structure... "co-located", or "equal-order"



- We have the choice of where the DOFs are solved... Our choice?

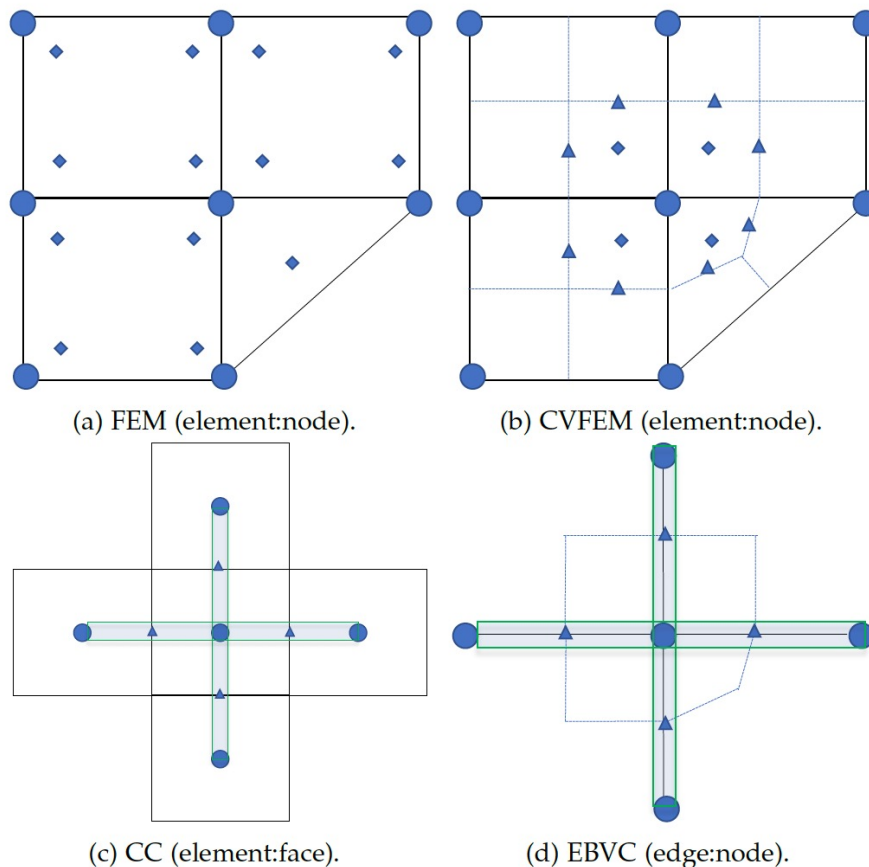


Figure 6: Patches of elements that define center degree-of-freedom stencil.

- Here, all variables are solved at the same location (nodes/vertices or cell-centers)
- Underlying basis for each DOF is the same, i.e., "equal-order" interpolation
- We already have noted the gradient operator in 1-D that might pose some problems...

$$G_x p = \frac{(p_{j+1} - p_{j-1})A_x}{2}$$

$$G_x p = M^{-1} \frac{(p_{j+1} - p_{j-1})A_x}{2}$$

Stencil for CC-quantities ●



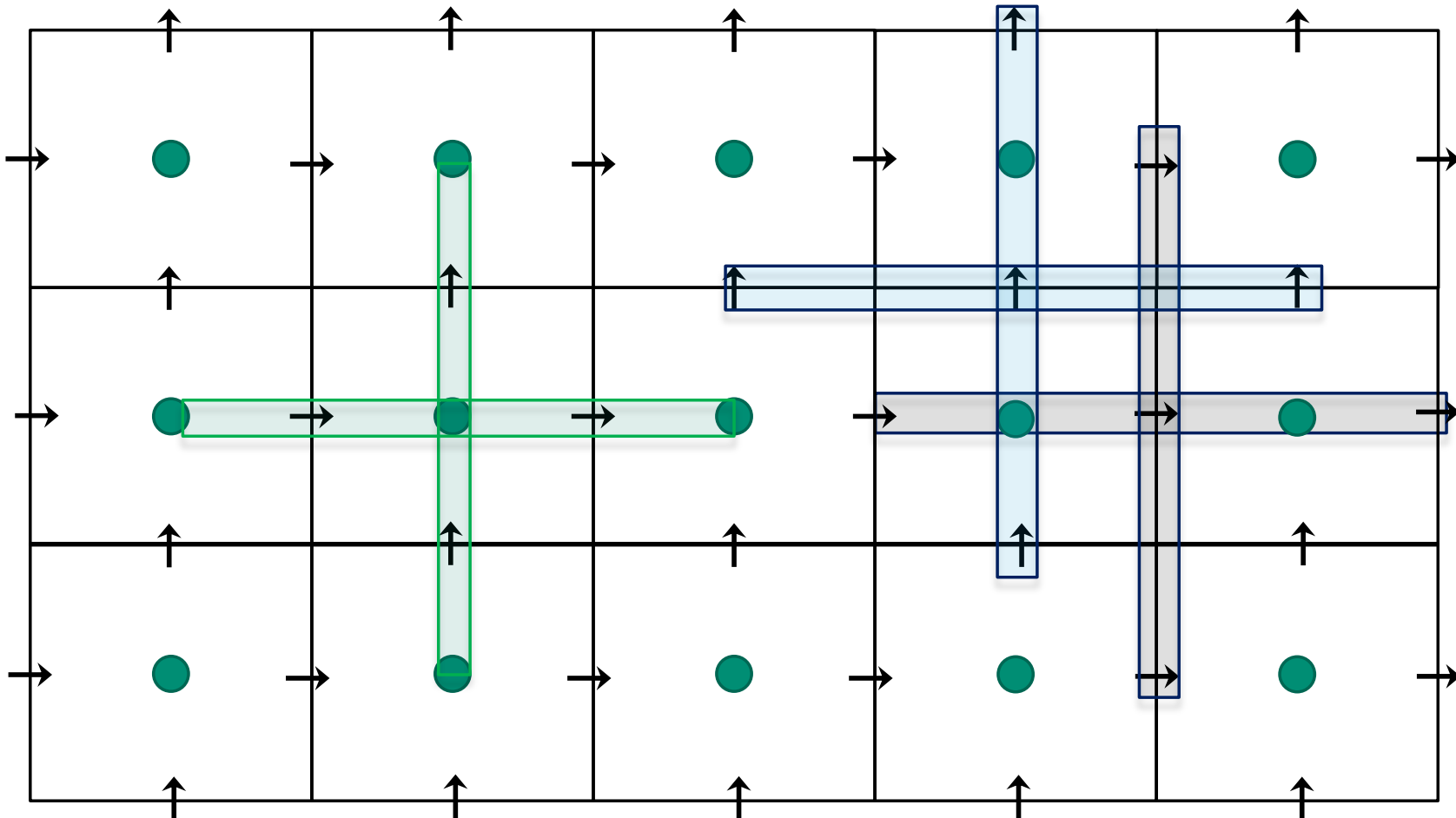
Stencil for x-velocity →



Stencil for y-velocity ↑



- Velocity degree-of-freedom is staggered relative to pressure and other primitives, e.g., enthalpy, mixture fraction, etc.



Staggered.....

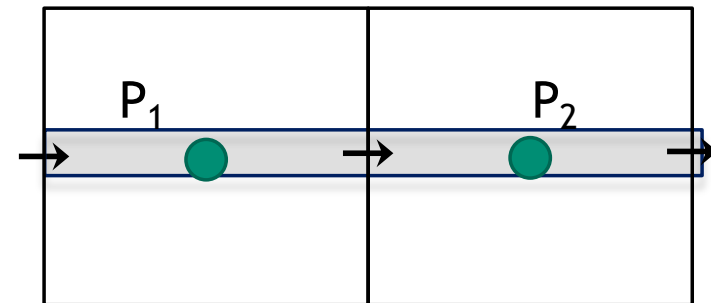
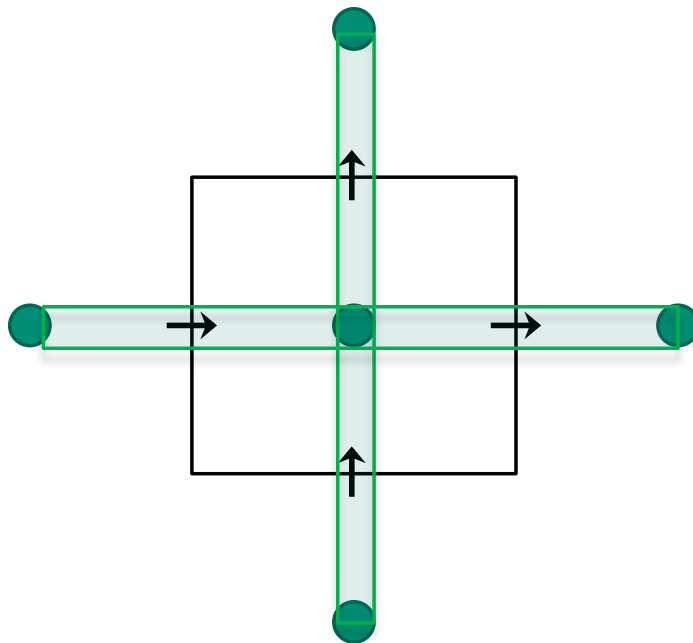
Stencil for CC-quantities ●



Stencil for x-velocity →



- Operators are now very compact and local...



$$G_x p = M^{-1}(p_2 - p_1)A_x$$

Incremental Pressure-Projection without Pressure Stabilization



- Let the inverse of $\mathbf{A}$, $\mathbf{A}^{-1}$ be approximated by $\mathbf{B}_2$ as a scalar, τ
- Let $\mathbf{B}_1$ be equal to the scaled Laplace operator, $-\tau\mathbf{L}$

$$\text{Momentum and Continuity: } \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{D} & -\tau\mathbf{L} \end{bmatrix} \begin{bmatrix} \hat{u} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{f} \\ -\tau\mathbf{L}p^n \end{bmatrix} \left\{ \begin{array}{l} \mathbf{A}\hat{u} = \hat{f} = \mathbf{f} - \mathbf{G}p^n \\ \mathbf{D}\hat{u} - \tau\mathbf{L}(p^{n+1} - p^n) = 0 \end{array} \right.$$

$$\text{Nodal Projection: } \begin{bmatrix} \mathbf{I} & \tau\mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{u} \\ \hat{p} \end{bmatrix} + \begin{bmatrix} \tau\mathbf{G}p^n \\ 0 \end{bmatrix} \left\{ \begin{array}{l} u^{n+1} = \hat{u} - \tau\mathbf{G}(p^{n+1} - p^n) \\ p^{n+1} = \hat{p} \end{array} \right.$$

- The new splitting and stabilization error is given by:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} (\mathbf{I} - \mathbf{A}\tau)\mathbf{G}(p^{n+1} - p^n) \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})(p^{n+1} - p^n) \end{bmatrix}$$

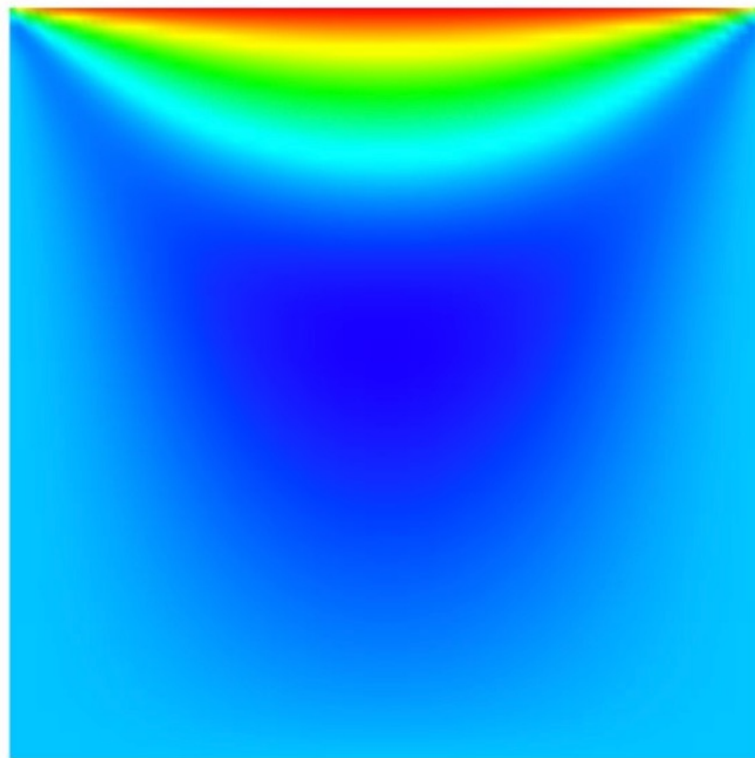
Examples:

- Dwyer (1990)
- Almgren (2000)

- The above can be shown to demonstrate second-order temporal error (coming)
- A scheme can be designed such that $\mathbf{L} = \mathbf{D}\mathbf{G}$ (staggered)
- A scheme in which $\mathbf{L} \neq \mathbf{D}\mathbf{G}$ (collocated or equal-order) can show that $\mathbf{L} - \mathbf{D}\mathbf{G} \sim 4^{\text{th}}$ -order pressure stabilization, however, in this case acting on $p^{n+1} - p^n$

The Role of Pressure Stabilization ($L \neq DG$) in an Equal-Order Approach

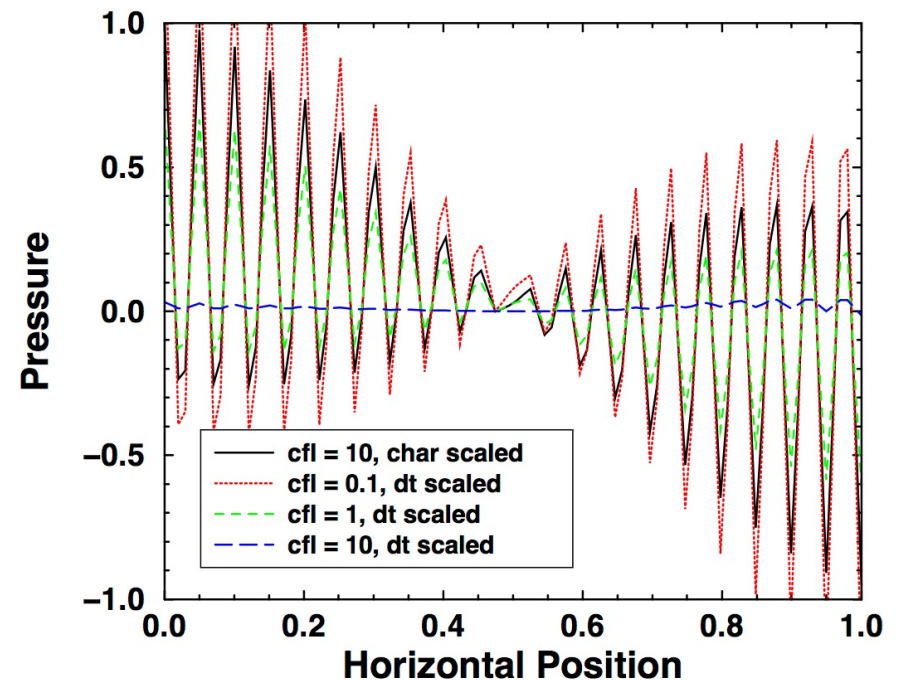
- Consider the classic lid-driven cavity flow with top wall velocity of U_0



lid-driven cavity velocity
(u-component)

u

9.813e-01
6.840e-01
3.867e-01
8.934e-02
-2.080e-01



Unsmoothed pressure field at
various Courant numbers

Equal-Order: same basis and interpolation operators for continuity and momentum
Also known as: “collocated”

Incremental Approximate Pressure-Projection with Pressure Stabilization Errors



- Let the inverse of $\mathbf{A}$, $\mathbf{A}^{-1}$ be approximated by $\mathbf{B}_2$ as a scalar, τ (which is $\sim$ time scale)
- Let $\mathbf{B}_1$ be equal to the scaled Laplace operator, $-\tau \mathbf{L}$

$$\text{Momentum and Continuity: } \begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & -\tau \mathbf{L} \end{bmatrix} \begin{bmatrix} \hat{u} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{f} \\ -\mathbf{D}\tau \mathbf{G}p^n \end{bmatrix} \left\{ \begin{array}{l} \mathbf{A}\hat{u} = \mathbf{f} - \mathbf{G}p^n \\ \mathbf{D}\hat{u} = \tau(\mathbf{L}p^{n+1} - \mathbf{D}\mathbf{G}p^n) \end{array} \right.$$

$$\text{Nodal Projection: } \begin{bmatrix} \mathbf{I} & \tau \mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{u} \\ \hat{p} \end{bmatrix} + \begin{bmatrix} \tau \mathbf{G}p^n \\ 0 \end{bmatrix} \left\{ \begin{array}{l} u^{n+1} = \hat{u} - \tau \mathbf{G}(p^{n+1} - p^n) \\ p^{n+1} = \hat{p} \end{array} \right.$$

- The new splitting and stabilization error is given by:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} (\mathbf{I} - \mathbf{A}\tau)\mathbf{G}(p^{n+1} - p^n) \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})p^{n+1} \end{bmatrix}$$

Examples:

- Rhie-Chow (1983)
- Peric (1985)
- Domino (2006)

- The above can be shown to hold a second-order temporal error (coming)
- Here, due to equal-order interpolation, i.e., collocation of primitives, $\mathbf{L} \neq \mathbf{D}\mathbf{G}$
- Therefore, $\mathbf{L} - \mathbf{D}\mathbf{G} \sim 4^{\text{th}}$ -order pressure stabilization (pressure oscillations damped)
- However, pressure-stabilization error remains



Non-Incremental Pressure-Projection: “Pressure-Free” with or without Pressure Stabilization

- Let the inverse of $\mathbf{A}$, $\mathbf{A}^{-1}$ be approximated by $\mathbf{B}_2$ as a scalar, τ
- Let $\mathbf{B}_1$ be equal to the scaled Laplace operator, $-\tau\mathbf{L}$

Momentum and Continuity:
$$\begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & -\tau\mathbf{L} \end{bmatrix} \begin{bmatrix} \hat{u} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} \quad \left\{ \begin{array}{l} \mathbf{A}\hat{u} = \mathbf{f} \\ \mathbf{D}\hat{u} - \tau\mathbf{L}(p^{n+1}) = 0 \end{array} \right.$$

Nodal Projection:
$$\begin{bmatrix} \mathbf{I} & \tau\mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{u} \\ \hat{p} \end{bmatrix} + \begin{bmatrix} \tau\mathbf{G}p^n \\ 0 \end{bmatrix} \quad \left\{ \begin{array}{l} u^{n+1} = \hat{u} - \tau\mathbf{G}(p^{n+1}) \\ p^{n+1} = \hat{p} \end{array} \right.$$

- The new splitting and stabilization error is given by:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} (\mathbf{I} - \mathbf{A}\tau)\mathbf{G}(p^{n+1}) \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})(p^{n+1}) \end{bmatrix}$$

Examples:

- Kim and Moin (1985)

- A fully-implicit scheme can be shown to demonstrate first-order temporal error (coming)
- A scheme can be designed such that $\mathbf{L} = \mathbf{D}\mathbf{G}$ (staggered) to remove stabilization error
- A scheme in which $\mathbf{L} \neq \mathbf{D}\mathbf{G}$ (collocated or equal-order) can show that $\mathbf{L}-\mathbf{D}\mathbf{G} \sim 4^{\text{th}}$ -order pressure stabilization (however, in this case acting on $p^{n+1}-p^n$)

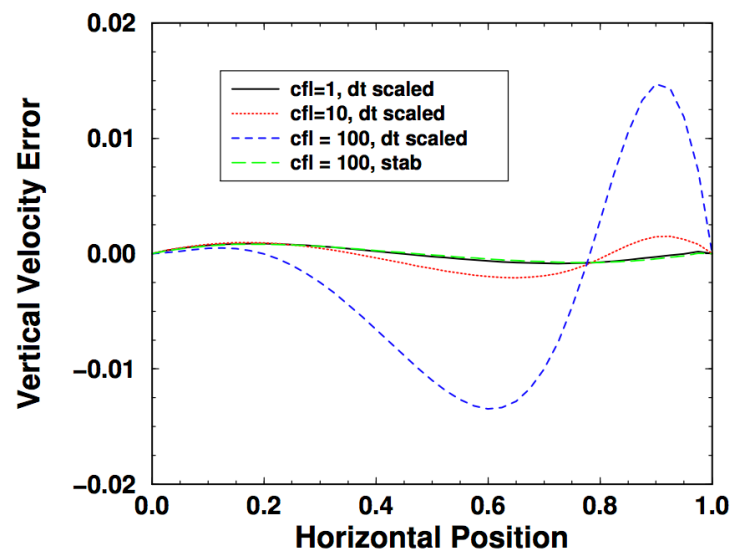
Sensitivity to chosen τ

- Recall that the equal-order pressure stabilization error is given by,

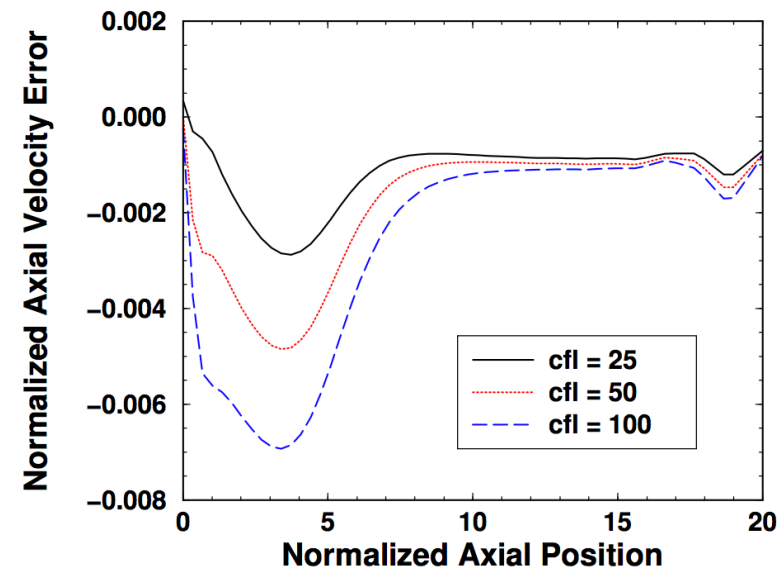
$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} (\mathbf{I} - \mathbf{A}\tau)\mathbf{G}(p^{n+1} - p^n) \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})p^{n+1} \end{bmatrix}$$

Stabilizing effect

- Practical examples of error as a function of:
 - $\tau \propto \Delta t$ (the simulation timestep)
 - $\tau^c \propto (u/\Delta x)^{-1}$ (characteristic advection time scale)
 - $\tau^c(\mathbf{L} - \mathbf{D}\mathbf{G})p^{n+1} + \Delta t\mathbf{L}(p^{n+1} - p^n)$ (Soto and Lohner, “stabilized”)



Driven Cavity



Open Jet

Monolithic Staggered or Equal-order Interpolation (Collocated)

- Note that we need not split the system for a staggered scheme:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & -\tau \mathbf{L} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ -\tau \mathbf{L} p^n \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})(p^{n+1} - p^n) \end{bmatrix}$$

- or collocated:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & -\tau \mathbf{L} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ -\tau \mathbf{D}\mathbf{G} p^n \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})p^{n+1} \end{bmatrix}$$

- Conclusion: Monolithic schemes control splitting error, however, dealing with pressure stabilization is an additional complexity for equal-order methods regardless of the chosen approach to solve the coupled system



- In Pressure-Stabilized Petrov-Galerkin (PSPG) methods (Hughes et al, 1985):

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & -\tau_{PSPG} \mathbf{G} \mathbf{q} \cdot \mathbf{M} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- Here, $\mathbf{M}$ is a fine scale momentum residual and q is the test function for the Finite Element Method (continuity equation)
- Note that $\mathbf{M}$ contains a local pressure gradient which, thereby, provides the pressure stabilization
- The fine-scale momentum residual is evaluated locally at the quadrature point and with mesh refinement reduces at a design-order rate
- With some algebra, one can show that $\mathbf{L-DG} \sim \mathbf{M}$
- For more references, see Majumdar, Numerical Heat Transfer, 1988; Tezduyar and Sathe, J. Comput. Appl. Mech., 2003
- Here, τ can be:
 - Simulation time step, i.e., Δt
 - Local advection/diffusion time scale, $\left(\frac{u}{\Delta x} + \frac{v}{\Delta x^2}\right)^{-1}$ - or a more accurate flow-aligned approach
 - Diagonal of A
 - Full inverse of A (Ozawa)

Code Verification To Establish Accuracy

- Consider a two-dimensional transient solution to the incompressible equations of motion:
- This solution is known as the convecting, decaying, Taylor vortex

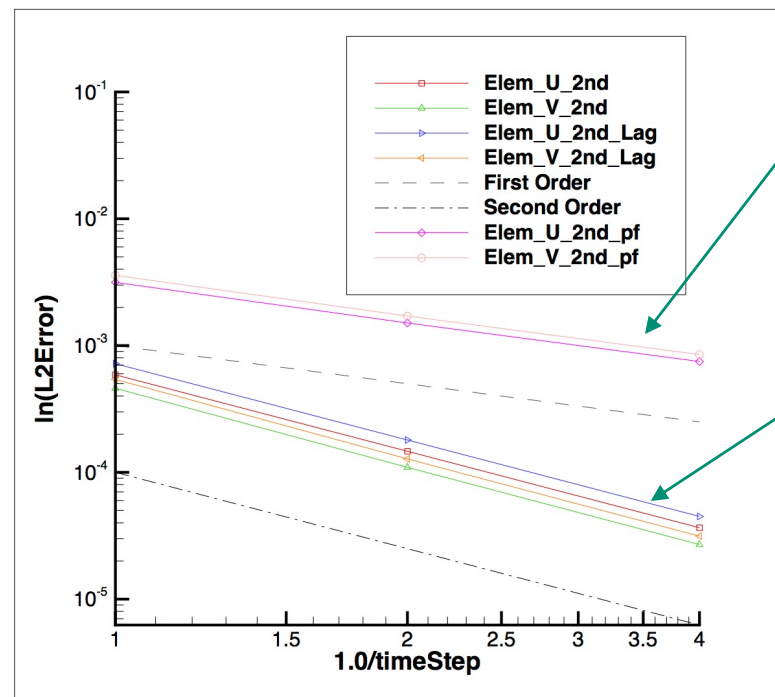
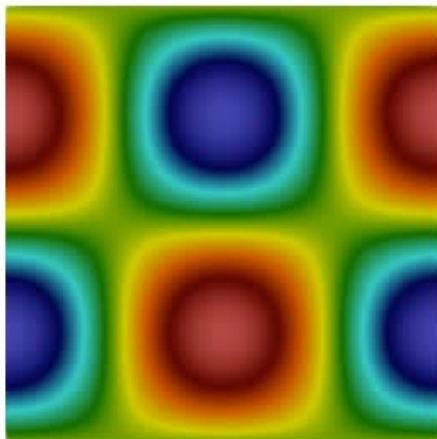
$$u = u^o - \cos(\pi(x - u^o t)) \sin(\pi(y - v^o t)) e^{-2\omega t}$$

$$v = v^o + \sin(\pi(x - u^o t)) \cos(\pi(y - v^o t)) e^{-2\omega t}$$

$$p = -\frac{p^o}{4} [(\cos(2\pi(x - u^o t)) + \cos(\pi(y - v^o t)))] e^{-4\omega t}$$

Time: 0.000000

velocity_X
1.1e-15 1 2.0e+00



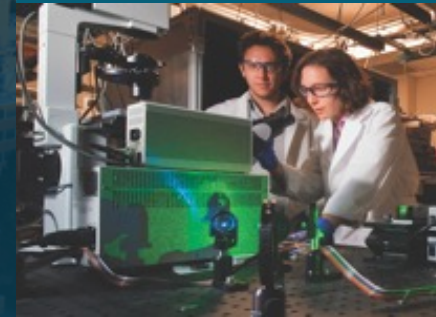
Approximate factorization

Flavors of incremental



- Block matrix and operator form represents a useful construct to analyze coupling and stabilization
- Approximate Factorization is generally $O(\Delta t)$
- With very simple modifications, splitting error is mitigated
- Detailed code verification is a critical tool to both test theoretical understandings in addition to establishing a proper code implementation

Stanford ME469: Splitting and Stabilization Errors Review



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- Consider the monolithic, uniform density, low-Mach equation system:

$$\frac{\partial u_j}{\partial x_j} = 0$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} = -\frac{\partial p}{\partial x_i} + S_i$$

that can be written in block form as:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- We seek to factorize this system via:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & \mathbf{B}_1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{B}_2 \mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \approx \begin{bmatrix} \mathbf{A} & \mathbf{A} \mathbf{B}_2 \mathbf{G} \\ \mathbf{D} & (\mathbf{B}_1 + \mathbf{D} \mathbf{B}_2 \mathbf{G}) \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

the exact factorization can be recovered by defining:

$$\left\{ \begin{array}{l} \mathbf{B}_2 = \mathbf{A}^{-1} \\ \mathbf{B}_1 = -\mathbf{D} \mathbf{B}_2 \mathbf{G} \end{array} \right.$$

- $\mathbf{B}_2$ determines the projection time scale, ideally chosen to approximate $\mathbf{A}^{-1}$
- $\mathbf{B}_1$ controls the projection error, ideally chosen to cancel $\mathbf{B} \mathbf{D}_2 \mathbf{G}$

Coupling Definitions: Monolithic

Fully Coupled, or Monolithic

- One equation system, one solve

For a given
time step:

$$\left. \begin{array}{l} \text{do while (!converged) \{ \\ \left[\begin{array}{cccc} \frac{\partial}{\partial p} C & \frac{\partial}{\partial \tilde{u}_x} C & \frac{\partial}{\partial \tilde{u}_y} C & \frac{\partial}{\partial \tilde{z}} C \\ \frac{\partial}{\partial p} \tilde{U}_x & \frac{\partial}{\partial \tilde{u}_x} \tilde{U}_x & \frac{\partial}{\partial \tilde{u}_y} \tilde{U}_x & \frac{\partial}{\partial \tilde{z}} \tilde{U}_x \\ \frac{\partial}{\partial p} \tilde{U}_y & \frac{\partial}{\partial \tilde{u}_x} \tilde{U}_y & \frac{\partial}{\partial \tilde{u}_y} \tilde{U}_y & \frac{\partial}{\partial \tilde{z}} \tilde{U}_y \\ \frac{\partial}{\partial p} \tilde{Z} & \frac{\partial}{\partial \tilde{u}_x} \tilde{Z} & \frac{\partial}{\partial \tilde{u}_y} \tilde{Z} & \frac{\partial}{\partial \tilde{z}} \tilde{Z} \end{array} \right] \begin{bmatrix} \Delta p \\ \Delta \tilde{u}_x \\ \Delta \tilde{u}_y \\ \Delta \tilde{z} \end{bmatrix} = - \begin{bmatrix} resC \\ res\tilde{U}_x \\ res\tilde{U}_y \\ res\tilde{Z} \end{bmatrix} \end{array} \right\}$$

- Reduced radius of convergence
- Second order convergence

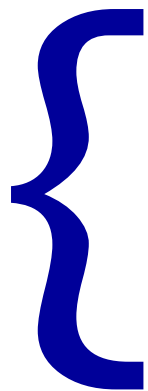
Coupling Definitions: Segregated

Loosely Coupled, or Segregated

- Multiple matrix systems solved within a Picard linearization, i.e., Jacobi or Gauss-Seidel iteration. Coupling is two-way (or multiple way if the physics becomes really complex)

do while (!converged) {

For a given
time step:



$$\begin{aligned}
 &\bar{\rho} = f(\tilde{Z}) \\
 &\left[\frac{\partial}{\partial \tilde{u}_x} \tilde{U}_x \right] [\Delta \tilde{u}_x] = -[res \tilde{U}_x] \\
 &\left[\frac{\partial}{\partial \tilde{u}_y} \tilde{U}_y \right] [\Delta \tilde{u}_y] = -[res \tilde{U}_y] \\
 &\tilde{u}_i^{k+1} = \tilde{u}_i^k + \alpha \Delta \tilde{u}_i \\
 &\left[\frac{\partial}{\partial p} C \right] [\Delta p] = -[res P] \\
 &\tilde{u}_i^{n+1} = \tilde{u}_i^{k+1} - \tau \nabla (\Delta p^{n+1/2}) \\
 &\left[\frac{\partial}{\partial \tilde{Z}} \tilde{Z} \right] [\Delta \tilde{Z}] = -[res \tilde{Z}] \\
 &\tilde{Z}^{k+1} = \tilde{Z}^k + \alpha \Delta \tilde{Z}
 \end{aligned}$$

}

Every equation is segregated..

- Larger radius of convergence
- Often, stagnates to first-order convergence



Incremental Approximate Pressure-Projection with Pressure Stabilization Errors

- Let the inverse of $\mathbf{A}$, $\mathbf{A}^{-1}$ be approximated by $\mathbf{B}_2$ as a scalar, τ (which is $\sim$ time scale)
- Let $\mathbf{B}_1$ be equal to the scaled Laplace operator, $-\tau \mathbf{L}$

$$\text{Momentum and Continuity: } \begin{bmatrix} \mathbf{A} & 0 \\ \mathbf{D} & -\tau \mathbf{L} \end{bmatrix} \begin{bmatrix} \hat{u} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{f} \\ -\mathbf{D}\tau \mathbf{G}p^n \end{bmatrix} \left\{ \begin{array}{l} \mathbf{A}\hat{u} = \mathbf{f} - \mathbf{G}p^n \\ \mathbf{D}\hat{u} = \tau(\mathbf{L}p^{n+1} - \mathbf{D}\mathbf{G}p^n) \end{array} \right.$$

$$\text{Nodal Projection: } \begin{bmatrix} \mathbf{I} & \tau \mathbf{G} \\ 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \hat{u} \\ \hat{p} \end{bmatrix} + \begin{bmatrix} \tau \mathbf{G}p^n \\ 0 \end{bmatrix} \left\{ \begin{array}{l} u^{n+1} = \hat{u} - \tau \mathbf{G}(p^{n+1} - p^n) \\ p^{n+1} = \hat{p} \end{array} \right.$$

- The new splitting and stabilization error is given by:

$$\begin{bmatrix} \mathbf{A} & \mathbf{G} \\ \mathbf{D} & 0 \end{bmatrix} \begin{bmatrix} u^{n+1} \\ p^{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} + \begin{bmatrix} (\mathbf{I} - \mathbf{A}\tau)\mathbf{G}(p^{n+1} - p^n) \\ \tau(\mathbf{L} - \mathbf{D}\mathbf{G})p^{n+1} \end{bmatrix}$$

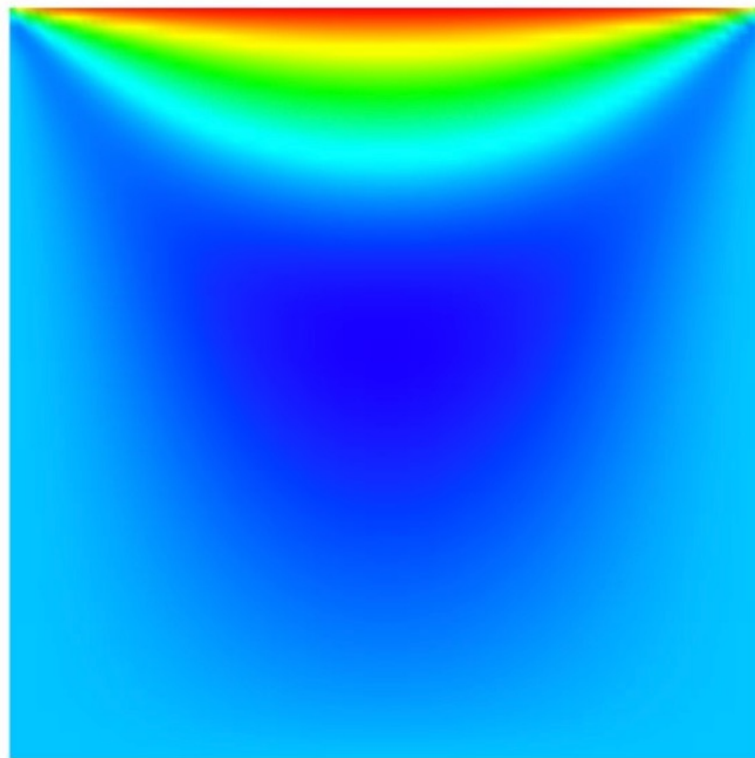
Examples:

- Rhie-Chow (1983)
- Peric (1985)
- Domino (2006)

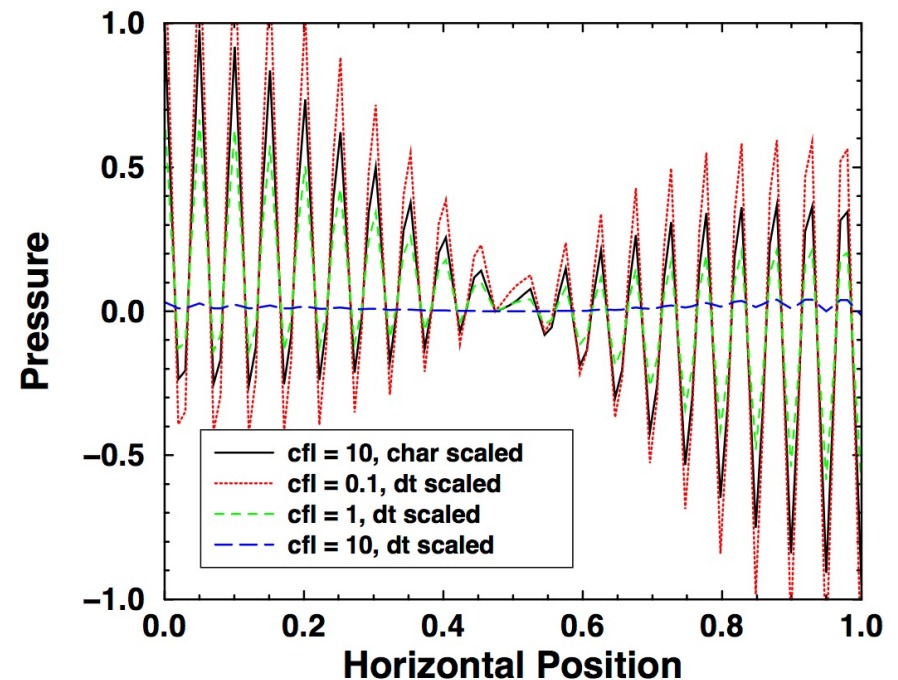
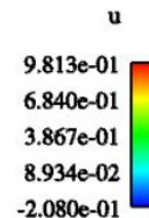
- The above can be shown to hold a second-order temporal error (coming)
- Here, due to equal-order interpolation, i.e., collocation of primitives, $\mathbf{L} \neq \mathbf{D}\mathbf{G}$
- Therefore, $\mathbf{L} - \mathbf{D}\mathbf{G} \sim 4^{\text{th}}$ -order pressure stabilization (pressure oscillations damped)
- However, pressure-stabilization error remains

The Role of Pressure Stabilization ($L \neq DG$) in an Equal-Order Approach

- Consider the classic lid-driven cavity flow with top wall velocity of U_0



lid-driven cavity velocity
(u-component)



Unsmoothed pressure field at
various Courant numbers

Equal-Order: same basis and interpolation operators for continuity and momentum
Also known as: “collocated”

First, The Role of $\dot{m}$

- For an equal-order, low-Mach approximate projection scheme, explicit pressure stabilization was added. How does this manifest itself in the advection operator?

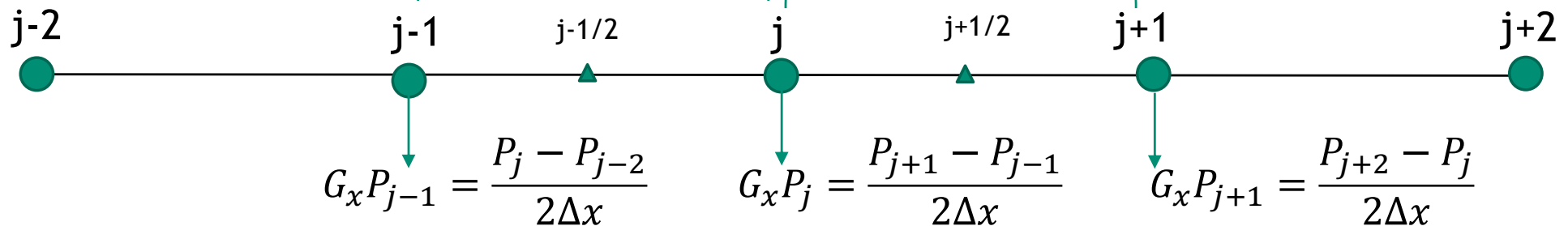
Recall discrete continuity solve

$$\int w \frac{\partial \rho u_j Z}{\partial x_j} dV = \int \rho u_j Z n_j dS \approx \sum_{ip} \dot{m} Z_{ip}$$

$$\mathbf{D}\hat{u} = \tau(\mathbf{L}p^{n+1} - \mathbf{D}\mathbf{G}p^n)$$

$$\frac{\partial P_{j-1/2}}{\partial x} = \frac{P_j - P_{j-1}}{\Delta x}$$

$$\frac{\partial P_{j+1/2}}{\partial x} = \frac{P_{j+1} - P_j}{\Delta x}$$



- Using the above equations, we can derive the actual continuity equation that we are solving:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} \neq 0 \propto \tau \frac{\partial^4 P}{\partial x_j^4} \Delta x_j^3$$

Wait, What is an alternative view of what L-DG forms?

- Let's define an assembled system for velocity component as:

$$\mathbf{T}u_i + \mathbf{V}u_i - \mathbf{D}u_i - \mathbf{S} = -\mathbf{G}p_i$$

Where $\mathbf{T}$, $\mathbf{V}$, $\mathbf{D}$, $\mathbf{S}$, and $\mathbf{G}$ are the time, advection, diffusion, source and gradient operators (matrix form)

- We can do the same for a local integration point fine-scale momentum equation:

$$Tu_i + Vu_i - Du_i - S = -dp/dx_i|_{ip}$$

Algebraically, we can reconstruct the fine scale residual to be an interpolation of the nodal-assembled momentum equation that we solved:

- $(dp/dx_i - \mathbf{M}^{-1}\mathbf{G}p_i|_{ip}) = \text{Res}_{u_i}$

.... And uses this scaled residual as a correction to the interpolated velocity

- The finite volume community calls this “Rhie-Chow”, or “Momentum interpolation”, while the finite element community terms this “residual-based stabilization”